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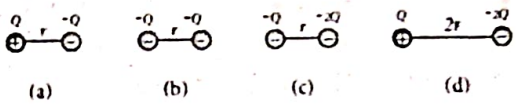
12 - STD *Quarterly Examination September -2025*

Time : 3:00 Hours **Physics** *Marks: 70*

I. Answer all the questions. 15X1=15

1. Rank the electrostatic potential energies for the given system of charges in increasing order

a)1 = 4 < 2 < 3 b) 2 = 4 < 3 < 1 c) 2 = 3 < 1 < 4 d) 3 < 1 < 2< 4



2. An electric field $\vec{E} = 10x\vec{\imath}$ exists in a certain region of space. Then the potential difference $V = V_o - V_A$, where V_o is the potential at the origin and V_A is the potential at x = 2 m is

a) 10 V b) -20 V c) +20 V d) -10 V

3. A wire of resistance 2 ohm per meter is bent to form a circle of radius 1m. The equivalent resistance between its two diametrically opposite points, A and B as shown in the figure is

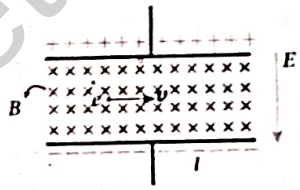


a) $\pi\Omega$ b) $\frac{\pi}{2}\ \Omega$ c) $2\pi\Omega$ d) $\frac{\pi}{4}\ \Omega$

4. In India electricity is supplied for domestic use at 220 V. It is supplied at 110 V in USA. If the resistance of a 60W bulb for use in India is R, the resistance of a 60W bulb for use in USA will be

a) R b) 2R c) R/4. d) R/2

5. An electron moves in a straight line inside a charged parallel plate capacitor of uniform charge density σ. The time taken by the electron to cross the parallel plate capacitor undeflected when the plates of the capacitor are kept under constant magnetic field of Induction($\bar{B}$) is



a) $\varepsilon_o \frac{elB}{\sigma}$ b) $\varepsilon_o \frac{lB}{\sigma l}$ c) $\varepsilon_o \frac{lB}{e\sigma}$ d) $\varepsilon_o \frac{lB}{\sigma}$

6. A non-conducting charged ring carrying a charge of q, mass m and radius r is rotated about its axis with constant angular speed ω. Find the ratio of its magnetic moment with angular momentum is

a) $\frac{q}{m}$ b) $\frac{2q}{m}$ c) $\frac{q}{2m}$ d) $\frac{q}{4m}$

7. When the current changes from +2A to −2A in 0.05 s, an emf of 8 V is induced in a coil. The co-efficient of self-induction of the coil is

a) 0.2 H b) 0.4 H c) 0.8 H d) 0.1 H

8. In a transformer, the number of turns in the primary and the secondary are 410 and 1230 respectively. If the current in primary is 6A, then that in the secondary coil is

a) 2A b) 18 A c) 12 A d) 1 A

9. A plane glass is placed over a various coloured letters (violet, green, yellow, red) The letter which appears to be raised more is,

(a) red (b) yellow (c) green (d) violet

10. In a Young's double-slit experiment, the slit separation is doubled. To maintain the same fringe spacing on the screen, the screen-to-slit distance D must be changed to,

a) 2D b) $\frac{D}{2}$ c) $\sqrt{2}\ D$ d) $\frac{D}{\sqrt{2}}$

11. The radius of curvature of curved surface at a thin planoconvex lens is 10 cm and the refractive index is 1.5. If the plane surface is silvered, then the focal length will be,

a) 5cm b) 10cm c) 15cm d) 20cm

12. Fraunhofer lines are an example of_____ spectrum.

(a) line emission (b) line absorption (c) band emission (d) band absorption

13.electromagnetic waves travel only in

a) Matter medium b) vacuum c) Oscillating electric and magnetic field d) Electric and magnetic fields oscillating in a direction perpendicular to each other

14.If a fish sees the sky without any obstruction, What is the total angle of view?. (refractive index of water $\sqrt{2}$)

a) 180^0 b) 90^0 c) 75^0 d) 60^0

15.If a metallic sphere without charge is brought near a positively charged sphere, then the sphere without charge

a) moves away from charged sphere b) moves towards the charged sphere c) Never deviate d) moves outwards or inwards depending on the magnitude of the charge.

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II. Answer any six questions. Question No. 24 is Compulsory. 6X2=12

16. What is interference of light?
17. Why does the sky appear blue?
18. What are polar molecules? Give an example.
19. Define temperature coefficient of resistance.
20. Find the power of a glass lens of focal length 150 cm .
21. State Flemings left hand rule.
22. What are step up and step down transformers?
23. What are the uses of infrared and ultraviolet rays?
24. Find the velocity of an electron moving in a region of an electric field E of magnitude 6.0×10^6 N C^{-1} and a magnetic field B of magnitude 0.83 T act perpendicular to each other.

III. Answer any six questions. Question No. 33 is Compulsory. 6X3=18

25. Derive an expression for motional emf from Lorentz force.
26. List the properties of electromagnetic waves (any six)
27. A rectangular sheet of sides 5 cm and 10 cm is placed across an uniform electric field of 100 N C^{-1} at an angle of 60^0 .Calculate the electric flux.If Θ is zero, then what is the value of electric flux?
28. State Kirchhoff 's current and voltage rule.
29. Derive an expression for the torque acting on a magnetic needle in a uniform magnetic field.
30. What are mirage and looming?
31. Derive the relation between phase difference and path difference.
32. Explain the working of lightning conductor.
33. The magnetic flux passes perpendicular to the plane of the circuit and is directed into the paper. If the magnetic flux varies with respect to time as per the following relation: $\Phi_B = (2t^3+3t^2+8t+5)$ Wb . what is the magnitude of the induced emf in the loop when t = 3 s?

IV. Answer all the questions. 5X5=25

34.(a) Find out the phase relationship between voltage and current in a pure inductive circuit.

or

(b) Discuss the working of cyclotron in detail..

35. (a)How is eddy current produced? How can it be minimised? Give its uses.

or

(b) Derive the equation for acceptance angle and numerical aperture of optical fibre.

36. (a)Explain the types of emission spectra

or

(b)Obtain an expression for the electrostatic potential due to an electric dipole.

37. (a)How are the emfs of two cells compared using potentiometer?

or

(b) Obtain an expression for the magnetic field at a point by a straight conductor of infinite length carrying current using Biot-Savart law.

38.(a)(i)Derive the effective resistance of resistors when they are connected in series.

(ii) Find the potential difference across 4 Ω and 6 Ω resistors connected in series with a 24 V battery. Also find the effective resistance in this circuit.

or

(b) (i)What is an optical path?

(ii)Light travels from air into a glass slab of refractive index 1.5 and 50 cm thickness.

(a) What is the speed of light in the glass slab?b) What is the time taken by light to pass through the glass slab?(c) Calculate the optical path of the glass slab.

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